Who benefits most?

Sorted partial effects · Interpretation, R code and inference

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Session overview

1. Literature and software.
2. A simulated treated/control example with age and sex.
3. Sorted effects and comparisons of the tails.
4. Inference and discussion.

Literature

01

The average effect is useful. What else can we learn from the same model?

↓ The method and its R implementation

Two starting points

- **Method:** Chernozhukov, Fernández-Val and Luo (2018), *Econometrica* (Chernozhukov, Fernández-Val, and Luo 2018).
- **Applied tutorial:** Chen et al. (2020), *The R Journal* (Chen et al. 2020).

Average effect

How large is the effect on average?

Sorted effects

How much do the fitted effects vary?

Classification

What characterizes the low and high tails?

What the R package provides

`spe()`

An estimated effect distribution and average effect, with **bootstrap confidence bounds**.

`ca()`

Comparisons of tail characteristics, with **bootstrap standard errors and joint p-values**.

We use **SortedEffects 1.7.0** for the main results.

Example

02

A binary outcome, two treatment groups, and two baseline characteristics.

↓ Define the effect before summarizing it

A simple simulated dataset

- **5,000 observations**, randomized to **treated** or **control**.
- Binary outcome: **1 = an adverse event**.
- Baseline variables: **age** (40–80 years) and **sex**.
- Treatment effects are allowed to vary with age and sex.

We will work with the reduction in event risk, in percentage points.

Simulated data throughout. The age and sex patterns illustrate the method; they are not empirical findings.

Predict both treatment conditions

Two ages, with sex held fixed (male).

Age 50

17.4% → 15.2% event risk
Control → treated

Age 70

26.8% → 11.4% event risk
Control → treated

Subtract the treated risk from the control risk.

The effect on the absolute scale

Risk reduction = control risk − treated risk.

Age 50

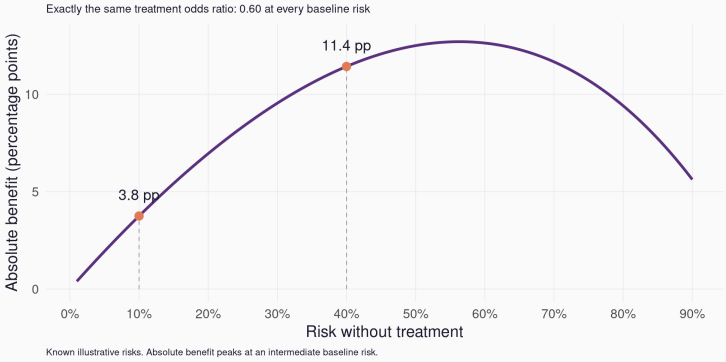
2.2 pp
17.4% → 15.2%

Age 70

15.4 pp
26.8% → 11.4%

- **Positive:** lower event risk under treatment.
- **Negative:** higher event risk under treatment.

Effect scale matters



- The treatment odds ratio is **0.60** at every baseline risk.
- Control risks of **10% and 40%** give risk reductions of **3.8 and 11.4 pp**.
- Absolute benefits can vary even without a treatment interaction on the log-odds scale.

Analytic conversion from odds to risks. The effect scale determines what counts as a constant effect.

What effect are we estimating?

$$\Delta(x) = \Pr(Y = 1 \mid A = 0, X = x) - \Pr(Y = 1 \mid A = 1, X = x).$$

- $A = 0$: control; $A = 1$: treated; X: age and sex.
- One **conditional average risk reduction** for each profile.
- The distribution of profiles defines the population we summarize.

Estimated effects

03

Fit the model, predict both conditions, then summarize the risk differences.

↓ R code and the sorted-effects curve

Fit the outcome model

```r
fm <- outcome ~ group * age_z * sex
fit <- glm(fm, data = dat, family = binomial())
```

- `group` has levels `control` and `treated`.
- `age_z = (age - 60) / 10`; `sex`: 0 = male, 1 = female.
- The interactions allow treatment effects to vary with age and sex.

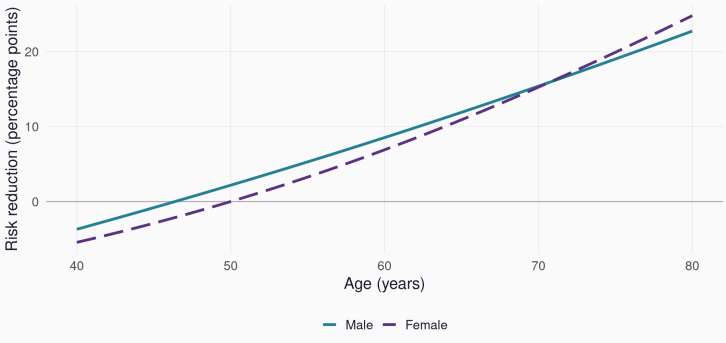
Predict the two risks

```r
control <- treated <- dat
control$group <- "control"
treated$group <- "treated"

benefit <- predict(fit, control, type = "response") -
  predict(fit, treated, type = "response")
```

- Keep age and sex fixed for each row.
- Change only the treatment assignment.
- `benefit` contains one predicted risk difference per row.

Before sorting: effects by age and sex



Simulated data. Fitted control risk minus fitted treated risk, at each age and sex.

- Each line shows fitted **risk reduction** at a given age and sex.
- Risk reduction rises with age in both simulated sex groups.
- Zero means equal predicted risks under treated and control.
- Next, combine all observed profiles and sort their effects.

Same model and scale as the following SPE curve. These demographic patterns are simulated.

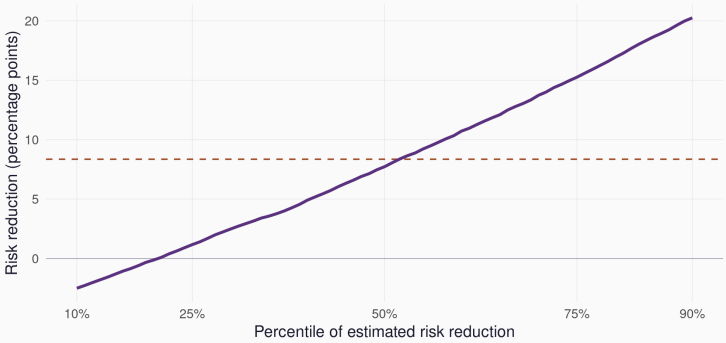
Calculate percentiles and the mean

```
u <- seq(.10, .90, by = .01)
spe_curve <- Hmisc::wtd.quantile(
  benefit, weights = rep(1, nrow(dat)), probs = u
)
mean(benefit)
```

- **SPE curve:** a percentile summary of the fitted effects.
- **Average effect:** the mean over all 5,000 profiles.

Equal profile weights. The weighted-quantile convention matches SortedEffects 1.7.0.

The sorted-effects curve



Simulated data. Dashed: average estimated risk reduction (8.4 pp).

- **Horizontal:** percentile of fitted risk reductions.
- **Vertical:** risk reduction in percentage points.
- Dashed line: mean of **8.4 pp**.
- Same model as the age–sex plot, evaluated at the **observed profiles** and sorted.

How to read one point

At the **80th percentile**, estimated risk reduction is **16.9 pp**.

- About **80% of profiles** have fitted reductions at or below this value.
- About **20%** have larger fitted reductions.
- This is a percentile **of effects**, not an effect on an outcome percentile.

It does not mean that an individual has an 80% chance of benefiting.

Classification

04

Which profiles have the smallest and largest estimated risk reductions?

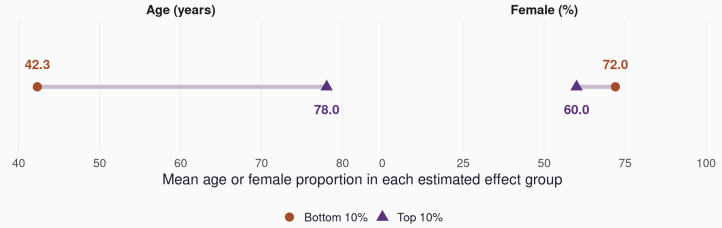
↓ Compare the bottom and top 10%

Describe the two tails

```r
cutoff <- Hmisc::wtd.quantile(
  benefit, rep(1, nrow(dat)), c(.10, .90)
)
tails <- list(low = benefit <= cutoff[1],
              high = benefit >= cutoff[2])
lapply(tails, function(i)
  colMeans(dat[i, c("age", "sex")]))
```

- Select groups using **estimated effects**, then summarize their covariates.
- Each tail contains about 500 observations.
- Both tails can include observations from either treatment arm.

Comparing the tails with dumbbells



Groups are defined by estimated risk reduction, not treatment assignment. The connecting line is not a confidence interval.

- Endpoints: means in the **bottom and top 10%** of fitted effects.
- Mean age: **42 vs 78 years**. Sex composition is less clearly separated.
- The line shows a group difference, **not a confidence interval**.
- These are descriptions of the fitted model's tails.

Inference

05

How much could the curve and group comparisons change in another sample?

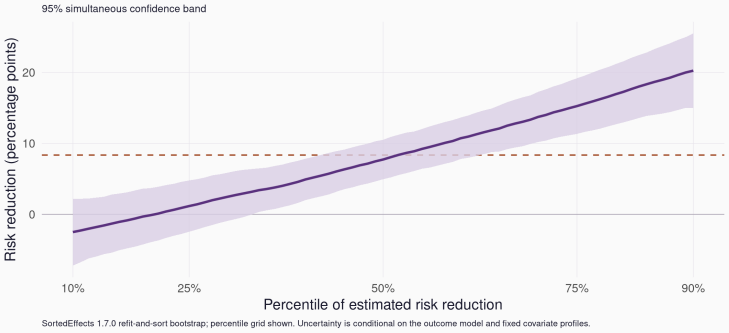
↓ Bootstrap inference in SortedEffects

Obtain a confidence band with `spe()`

```r
spe_fit <- SortedEffects::spe(
  fm = fm, data = dat, method = "logit",
  var_type = "categorical", var = "group",
  compare = c("treated", "control"),
  us = u,
  b = 499, alpha = .05, bc = FALSE, seed = 420
)
```

- Returns the curve, average effect and **bootstrap uncertainty**.
- This order of `compare` gives **control minus treated**.

Uncertainty in the sorted curve



- **95% uniform band** over the displayed grid.
- Bootstrap: **refit**, predict both risks, subtract and **sort**.
- Lower tail includes zero; upper tail stays positive.
- Uncertainty in the **curve**, not an individual effect.

SortedEffects 1.7.0; 499 draws; 10th–90th percentiles; uncorrected estimates. Observed age/sex profiles are held fixed for this SPE band.

Compare tail characteristics with `ca()`

```
ca_fit <- SortedEffects::ca(
  fm = fm, data = dat, method = "logit",
  var_type = "categorical", var = "group",
  compare = c("treated", "control"),
  u = .10, interest = "moment", cl = "diff",
  t = c("age", "sex"),
  b = 499, alpha = .05, bc = FALSE, seed = 420
)
```

Compares **mean age** and **female proportion** between the top and bottom tails.

Uncertainty in the tail differences

Characteristic	Top minus bottom	Bootstrap SE	Joint p
Age (years)	35.73	1.27	<0.01
Female (percentage points)	-12.00	31.53	0.88

- **Age:** a large, well-supported difference in this simulation.
- **Sex:** the apparent gap is uncertain after refitting and reclassification.
- Joint p-values adjust across the **two requested characteristics**.

Native `ca()` output; 499 draws. Displayed p < .01 is rounded Monte Carlo output. These are comparisons of tail profiles, not subgroup treatment-effect tests.

What the bootstrap does not settle

- **Identification:** is a causal interpretation justified by the design?
- **Model:** are conditional risks and interactions estimated well?
- **Population:** are these the profiles we want to summarize?
- **Analysis choices:** how sensitive are results to the model and tail cutoffs?

Discussion

06

When would this be more useful than an interaction plot or a subgroup table?

↓ Applications and limitations

What sorted effects add

- A summary of the **spread** of fitted effects, alongside the average.
- A way to compare tails without defining every subgroup in advance.
- Descriptions of the **profiles** in those tails, with inference.

Using the ranking to choose treatment needs a separate evaluation of benefits, harms and decision rules.

Questions for discussion

1. Where would a distribution of effects help in our own work?
2. What would SPE add to the analyses we already use?
3. Which model checks and sensitivity analyses would we require?
4. Would the bottom and top 10% be useful groups for that application?

Use the map to return to the age-by-sex plot, SPE curve, dumbbells or inference table.

Reading and reproducible code

- **Preparation:** Chen et al. (2020), *SortedEffects: Sorted Causal Effects in R* (Chen et al. 2020). Introduction and worked example.
- **Optional theory:** Chernozhukov, Fernández-Val and Luo (2018) (Chernozhukov, Fernández-Val, and Luo 2018).
- **Optional application (suggested):** Bilancini et al. (2022), blood-donor recruitment and retention during COVID-19 (Bilancini et al. 2022).
- **Repository:** slides, executable R code, results and checks at github.com/valentinklotzbuecher/jcod.

Additional material

+

Definitions, sensitivity diagnostics and software checks.

↓ Optional slides for questions

Backup · What is being sorted?

$$\mathsf{SPE}(u) = Q_{\Delta(X)}(u), \qquad \mathsf{APE} = E[\Delta(X)].$$

- The distribution of X defines the population being summarized.
- Classification describes baseline characteristics in selected effect tails.
- A quantile of effects differs from a difference of outcome quantiles.

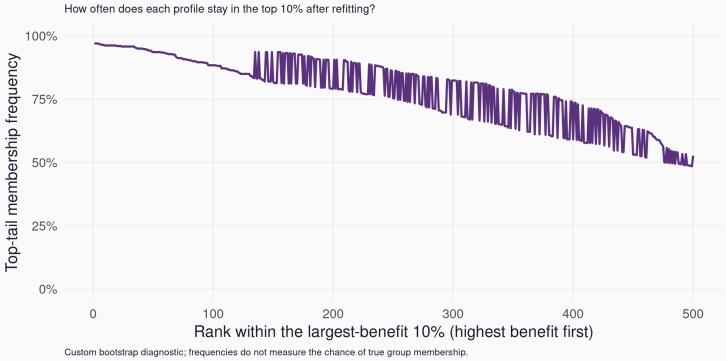
(Chernozhukov, Fernández-Val, and Luo 2018). Estimates use the package's weighted empirical quantile convention.

Backup · Simulation assumptions

- Baseline event risk depends on age, sex and their interaction.
- The treatment log odds ratio also depends on age, sex and their interaction.
- The fitted model matches the data-generating specification.
- The 5,000 observed profiles define the primary SPE reference distribution.

R/demo.R contains coefficients, seed and diagnostics. A matched-average constant-OR scenario is retained for software exercises.

Backup · Tail membership after refitting



- Each position is a profile in the original top 10%.
- Vertical position: how often it stays in the top 10% after refitting.
- Profiles near a cutoff can change groups.
- Frequencies are sensitivity summaries, not membership probabilities.

Separate teaching bootstrap, distinct from formal package classification inference.

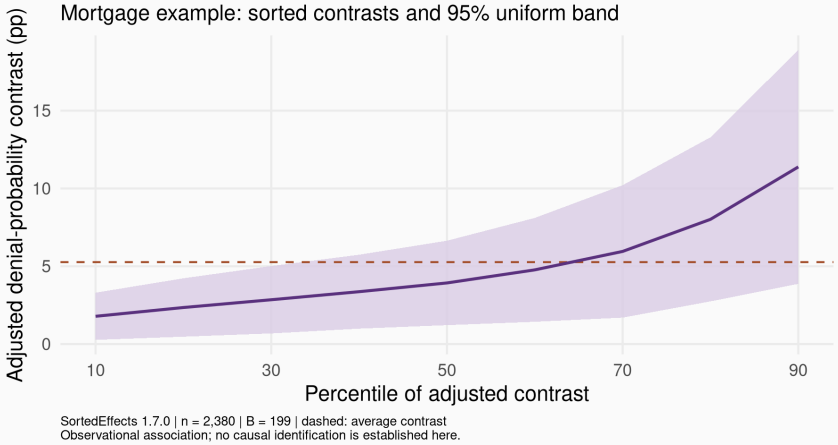
Backup · What is resampled?

- `spe()` **full sample:** refits with bootstrap weights; effect quantiles retain original profile weights.
- `ca()` **moments:** refits and updates thresholds and characteristic means with resampled weights.
- **Membership diagnostic:** separate fixed-profile refit-and-reclassify calculation.

Backup · Interpreting `ca()` output

- `interest = "moment", cl = "diff"` compares high- minus low-effect characteristic means.
- Joint p-values adjust across the **specified characteristics**.
- Bootstrap SEs and joint p-values are returned for these differences.
- Native moment confidence intervals are not returned by this call.

Backup · Original published example



Mortgage data from the tutorial (Munnell et al. 1996; Chen et al. 2020). Separate original-package benchmark.

References

Bilancini, and others, “Prosocial behavior in emergencies: Evidence from blood donors recruitment and retention during the COVID-19 pandemic,” *Social Science & Medicine*, 314 (2022), 115438.

Chen, Shuowen, Victor Chernozhukov, Iván Fernández-Val, and Ye Luo, “SortedEffects: Sorted causal effects in R,” *The R Journal*, 12 (2020), 131–146.

Chernozhukov, Victor, Iván Fernández-Val, and Ye Luo, “The sorted effects method: Discovering heterogeneous effects beyond their averages,” *Econometrica*, 86 (2018), 1911–1938.

Munnell, Alicia H., Geoffrey M. B. Tootell, Lynn E. Browne, and James McEneaney, “Mortgage lending in Boston: Interpreting HMDA data,” *The American Economic Review*, 86 (1996), 25–53.